Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science & Engineering

End Semester Test (EST)

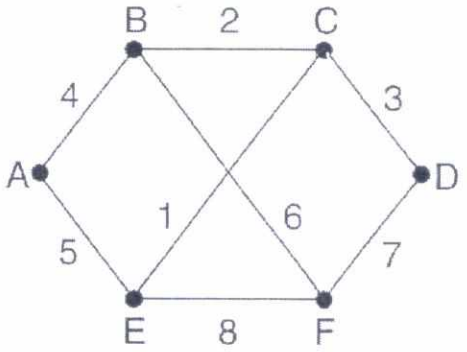
B.E: 3rd Year | Course Code: UCS520

Dec 18, 2025 | Course Name: Computer Network

M. Marks: 60 (Weightage:40) | Name of Faculty: SHD, GVR, AMR, SSH, MCL

Note: Attempt any FOUR questions. Attempt all parts of one question at one place. Assume missing data if any.

Q. 1 (a)	Discuss importance of subnetting. Assume an organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets. Solve the following queries: i. Find the subnet mask. ii. Find the number of addresses in each subnet. iii. Find the first and last addresses in subnet 1. iv. Find the first and last addresses in subnet 1024.	CO3, L3 (5)
(b)	Explain why a medium-size or large-size company does not want a block of class C addresses. Additionally, assume a service provider is granted a block of addresses starting with 150.80.0.0/16. The service provider wants to distribute these blocks to 2600 customers as follows. i. The first group has 200 medium-size businesses; each need 128 addresses. ii. The second group has 400 small businesses; each need 16 addresses. iii. The third group has 2000 households; each need 4 addresses. Design the sub-blocks and give the CIDR notation for each sub-block. Find out how many addresses are still available after these allocations.	CO3, L4 (4)
(c)	Draw and discuss the format of UDP datagram and TCP segment along with their fields in detail and also explain the difference between UDP and TCP in terms of connection establishment and reliability. Why is UDP considered a lightweight protocol compared to TCP?	CO2, L3 (6)
Q. 2 (a)	A company needs to send different types of messages over its network: i. A personal message from one employee to another, ii. A software update that must reach every device in the network, and iii. A training video that should be delivered only to a selected group of employees. Identify which type of network communication (unicast, broadcast, multicast) is most suitable for each above-scenario and logically justify your selection based on how data is transmitted in each communication type.	CO1, L1 (6)
(b)	Given the following 16-bit hexadecimal data: **4500 0034 A6F2 4000 4011 0000 C0A8 0001 C0A8 0002**. Assume that the checksum field is initially set to zero. Identify the individual fields represented by the given data and calculate the IPV4 checksum at the sender side.	CO3, L2 (6)
(c)	A sliding window protocol is designed for a 1 Mbps point to point link to the moon which has a one-way latency (delay) of 1.25 secs. Assuming that each frame carries 1 KB of data, what is the minimum number of bits needed for the sequence number?	CO1,L2 (3)
Q. 3 (a)	i. Most IP datagram reassembly algorithms have a timer to avoid having a lost fragment tie up reassembly buffers forever. Suppose that a datagram is fragmented into four fragments. The first three fragments arrive, but the last one is delayed. Eventually, the timer goes off and the three fragments in the receiver's memory are discarded. A little later, the last fragment stumbles in. What should be done with it? ii. What ICMP error-reporting Time Exceeded message indicates about the condition of the IPv4 packet during transit. iii. Discuss the format of the IPv6 base header and the significance of extension headers.	CO2, L1 (1+1+3)
(b)	Write a short note on following with suitable diagram.	CO4,L3

	i. DNS and SMTP protocol ii. DHCP protocol	(6)
(c)	Enumerate four distinct delay components in packet-switched networks in brief and discuss the three-phase process used in virtual-circuit switching.	CO1,L2 (4)
Q. 4 (a)	A host with IP address 192.20.40.20 and MAC address B2:34:55:10:22:10 wants to send a packet to another host with IP address 192.20.40.25 and MAC address A4:6E: F4:59:83: AB, which is initially unknown to the first host. Both hosts are on the same Ethernet network. i. Explain how ARP allows a host to discover the MAC address of another host on the same network. Draw and label the ARP Request and Reply frames, clearly defining all fields and discuss how changes in IP or MAC addresses of either host would affect the ARP process. ii. Consider a network with 20 systems, including system A and system B. System A wants to send 10 datagrams to system B in one second. a. Explain the effect of sending broadcasts if ARP is not used. b. Explain how ARP reduces broadcast processing and improves efficiency.	CO2,L3 (5+2)
(b)	Distinguish between Multilevel TDM, Multiple-slot TDM and Pulse stuffed TDM. Illustrate your explanation with appropriate examples and diagrams.	CO1, L1 (5)
(c)	Four digital channels, each with data rate 2 kbps, are multiplexed using synchronous TDM by taking 1 bit from each channel per frame. Assume the unit is 1 bit. Now calculate the input bit duration of each channel (in ms), the output bit rate on the multiplexed link (in kbps), and the frame duration (in ms).	CO1, L3 (3)
Q. 5(a)	Distinguish between open and closed loop congestion control policies with suitable examples and also describe how the congestion control policy works in TCP.	CO4, L2 (5)
(b)	i. Consider the given network, Distance vector routing is used, and the following vectors have just come in to router C: from B: (5, 0, 8, 12, 6, 2); from D: (16, 12, 6, 0, 9, 10); and from E: (7, 6, 3, 9, 0, 4). The cost of the links from C to B, D, and E, are 6, 3, and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the cost.  ii. From above, If the link C–B fails, explain whether this network is vulnerable to the count-to-infinity problem. Illustrate with a brief sequence of routing updates at C showing how costs to A could grow. iii. If costs are 8-bit in a 50-router network and distance vectors are exchanged twice per second, with each router having 3 links, how much bandwidth per full-duplex line is consumed by the routing updates?	CO4, L5 (5+3+2)
	********** All the Best **********	